Crop Yield

Weather-Uncertainty-Aware Indian Crop Yield Prediction Using Climatic Disturbance Indicators

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Abstract

Agriculture in India is highly dependent on climatic conditions. Traditional crop yield prediction models generally rely on historical production, rainfall, and temperature. However, large-scale climate variability and extreme weather events significantly influence crop productivity.

This project proposes a Weather-Uncertainty-Aware Crop Yield Prediction System that integrates three major climatic disturbance indicators:

- ENSO (El Niño–Southern Oscillation)
- Western Disturbances
- Cloudbursts

The outputs of these three weather modules are converted into a common spatial-temporal feature layer and combined with conventional agricultural features. The integrated dataset is then used to train machine learning models such as Random Forest, XGBoost, CatBoost, and TabPFN to predict crop yield more accurately.

Chapter 1: Introduction

Crop yield prediction is an essential component of modern agriculture. Accurate prediction helps:

- Farmers plan cultivation.
- Government agencies estimate food production.
- Agricultural departments manage resources.
- Policymakers prepare for climate-related risks.

Conventional models use historical crop and weather information but often overlook large-scale climatic uncertainty. This project incorporates multiple weather-risk indicators into the prediction process to improve forecasting accuracy.

Chapter 2: Objectives

The primary objectives are:

- Predict crop yield using machine learning.
- Integrate ENSO, Western Disturbance, and Cloudburst information.
- Develop a common feature engineering framework.
- Compare baseline and weather-aware prediction models.
- Improve forecasting under uncertain climatic conditions.

Chapter 3: System Architecture

The system consists of four major modules:

1. Historical climate and crop data.
2. ENSO prediction module.
3. Western Disturbance prediction module.
4. Cloudburst prediction module.

The outputs from all three weather modules are merged into a common feature layer before training the crop yield prediction model. The architecture concludes with yield prediction, prediction intervals, and weather-risk attribution.

Chapter 4: ENSO Vertical

Objective

The objective of the ENSO module is to forecast future ENSO conditions and convert them into climate uncertainty features for crop prediction.

Data Sources

Potential variables include:

- Niño 3.4 SST anomaly
- Niño-3 Index
- Niño-4 Index
- Southern Oscillation Index
- Indian Ocean Dipole
- Sea-level pressure
- Pacific wind anomalies

Feature Engineering

Features generated include:

- Lag variables
- Rolling averages
- ENSO anomaly
- El Niño probability
- Neutral probability
- La Niña probability

Output Features

- ENSO predicted anomaly
- El Niño probability
- Neutral probability
- La Niña probability
- Forecast confidence

Evaluation Metrics

- MAE
- RMSE
- Correlation
- F1-score
- ROC-AUC

These steps are described in the ENSO workflow, which emphasizes forecasting Niño 3.4 anomalies and deriving ENSO-state probabilities before integrating them into the crop prediction pipeline.

Chapter 5: Western Disturbance Vertical

Objective

Estimate the occurrence, intensity, and agricultural impact of Western Disturbances during crop-growing seasons.

Inputs

- Geopotential height
- Pressure
- Wind
- Rainfall
- Temperature
- Humidity
- Event tracks
- Regional coordinates

Dataset

Weekly records include:

- Region
- Week
- WD occurrence
- Intensity
- Rainfall

Output Features

Seasonal aggregation produces:

- Expected WD count
- WD probability
- Mean intensity
- Maximum intensity
- Expected rainfall
- Temperature effect
- Uncertainty score

Evaluation

Classification metrics:

- Precision
- Recall
- F1-score
- ROC-AUC

Regression metrics:

- RMSE
- MAE
- R^2

The document recommends converting weekly event predictions into crop-season summaries before using them as features.

Chapter 6: Cloudburst Vertical

Objective

Estimate localized extreme rainfall events that can damage crops.

Inputs

- Hourly rainfall
- Daily rainfall
- Relative humidity
- Convective energy
- Wind convergence
- Surface temperature
- Elevation
- Latitude
- Longitude

Dataset

Each observation represents:

District × Day

Output Features

- Cloudburst probability
- Expected extreme rainfall days
- Maximum cloudburst probability
- Cloudburst severity
- Waterlogging risk

Evaluation

- Precision
- Recall
- F1-score
- PR-AUC
- Calibration
- False-negative rate

The cloudburst workflow highlights aggregation of daily or weekly predictions into crop-season risk features for use in the final model.

Chapter 7: Common Spatial–Temporal Feature Layer

The three weather modules operate at different temporal and spatial scales.

To integrate them, the project defines a common forecasting unit:

District × Crop × Season × Year

Each weather module converts its outputs into this format.

The final feature table combines:

- Historical crop information
- Weather variables
- ENSO features
- Western Disturbance features
- Cloudburst features
- Spatial information
- Crop information

This common layer enables all weather-risk features to be merged consistently with crop and weather data.

Chapter 8: Machine Learning Models

The following algorithms are suggested:

- Random Forest
- XGBoost
- CatBoost
- TabPFN

The workflow performs:

1. Feature quality checks.
2. Model training.
3. Crop yield prediction.
4. Prediction interval estimation.
5. Explainability and weather-risk attribution.

Chapter 9: Model Comparison Strategy

The report recommends an ablation study.

Model	Features Used
Baseline	Historical crop + weather
Model A	Baseline + ENSO
Model B	Baseline + Western Disturbance
Model C	Baseline + Cloudburst
Model D	Baseline + ENSO + WD
Model E	Baseline + ENSO + Cloudburst
Model F	Baseline + WD + Cloudburst
Full Model	Baseline + ENSO + WD + Cloudburst

This comparison helps identify the contribution of each weather-risk vertical to crop yield prediction.

Chapter 10: Team Responsibilities

Member	Responsibility
Member 1	ENSO vertical
Member 2	Western Disturbance vertical
Member 3	Cloudburst vertical
Member 4	Feature integration, common pipeline, machine learning model

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Chapter 11: Output Contract

Every weather module should produce a standardized output containing:

- Region ID
- Year
- Season
- Crop window start
- Crop window end
- Risk probability
- Expected intensity
- Uncertainty score
- Data availability flag

Additional module-specific features include:

ENSO

- ENSO El Niño probability
- ENSO La Niña probability
- ENSO predicted anomaly

Western Disturbance

- WD event probability
- WD expected count
- WD expected rainfall

Cloudburst

- Cloudburst probability
- Extreme rainfall days
- Cloudburst severity

Using this shared schema allows the three independently developed weather modules to be merged into a single crop-yield forecasting dataset.

Conclusion

This project presents a comprehensive weather-aware crop yield prediction framework by integrating conventional agricultural information with three complementary weather-risk indicators: ENSO, Western Disturbances, and Cloudbursts. The proposed common spatial-temporal feature layer enables seamless integration of these diverse data sources into a unified machine learning pipeline. By comparing baseline and weather-enhanced models, the framework aims to quantify the predictive value of each climatic disturbance and improve crop yield forecasting under uncertain weather conditions.